TOPIC 07

Standard Error

Measuring how precise a sample statistic is: formulas, behaviour, estimation and bootstrap

WHAT YOU WILL LEARN

- Define the standard error and explain how it differs from the standard deviation
- Compute SE for a mean, a proportion and a difference
- See how sample size and variability change precision, and why returns diminish
- Estimate SE when σ is unknown, and by the bootstrap
- Report SE and error bars correctly

1. What is the standard error?

DEFINITION **Standard error (SE)**

The **standard error** of a statistic is the **standard deviation of its sampling distribution**. For the sample mean, it is the standard deviation of the distribution formed by sample means (Topic 05).

$$\mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Like any standard deviation, it shows **variability**, here the variability of the *sample means* from sample to sample. A small SE means different samples would give similar means, so any one sample mean is a **precise** estimate of μ. A large SE means a single sample mean could be far from μ.

KEY IDEA **Why SE appears in nearly every test and interval**

SE tells us **how well we have approximated the true mean**. Every confidence interval is 'estimate ± (reliability factor) × SE' and every test statistic is 'distance from the claimed value measured in SEs'.

2. Standard deviation versus standard error

	Standard deviation (σ or s)	Standard error (SE)
Describes	Spread of **individual observations**	Spread of a **statistic** across samples
Question	How different are people from each other?	How far might my estimate be from the truth?
Formula (mean)	σ (population) or s (sample)	σ/√n, or estimated by s/√n
As n grows	Stays about the same (settles on σ)	**Shrinks** towards 0
Use it to	Describe the data	Judge the estimate, build intervals and tests

COMMON MISTAKE **Do not mix them up**

Heights of 10,000 adults might have s = 8 cm, but the mean height from that sample has SE = 8/√10,000 = 0.08 cm. The mean is estimated *far* more precisely than any individual's height is predicted.

3. Computing and estimating the SE of the mean

The formula σ/√n needs the population standard deviation σ, which is usually unknown. We substitute the sample standard deviation s:

$$\widehat{\mathrm{SE}}(\bar{x}) = \frac{s}{\sqrt{n}} \qquad \text{where} \qquad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Software calls this SEM (standard error of the mean). When n is small and σ is replaced by s, the standardized mean follows a t distribution rather than the normal (Topic 09).

WORKED EXAMPLE **Data scientist salaries**

Salaries are normally distributed with a known σ = $15,000. A sample of n = 30 is taken. SE = 15,000/√30 = 15,000/5.477 = **$2,739** (about 2,739). A sample mean of this kind typically lands within roughly $2,739 of the true average, and within about $5,477 about 95% of the time.

WORKED EXAMPLE **Using s**

Eight measured values: 12.1, 14.3, 11.8, 15.2, 13.7, 12.9, 14.8, 13.1. x̄ = 13.488, s = 1.232, so SE = 1.232/√8 = **0.436**.

4. How sample size changes the SE

Because n sits under a square root, precision improves with n, but slowly:

n	SE = 15 / √n	Relative to n = 10
10	4.74	1.00×
30	2.74	0.58×
100	1.50	0.32×
400	0.75	0.16×
1600	0.38	0.08×

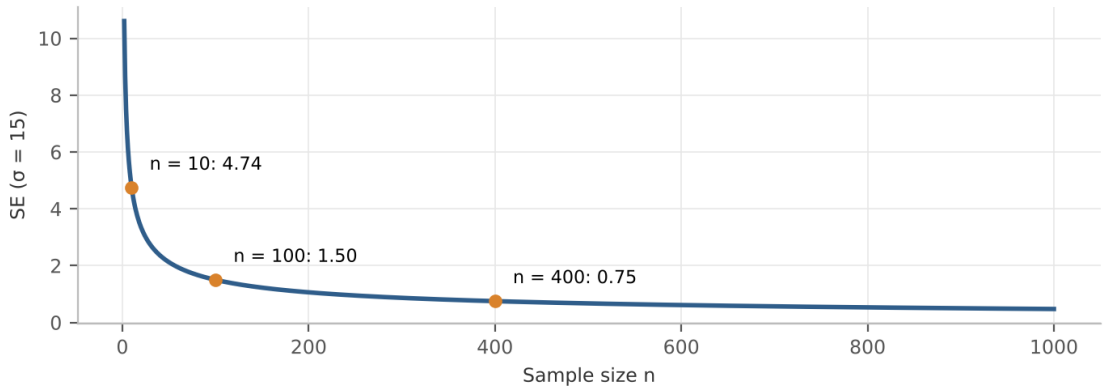


Figure 1. The SE falls steeply at first, then flattens. Going from 100 to 400 observations halves the SE, but moving from 400 to 700 helps far less.

KEY IDEA **The square-root law**

To **halve** the standard error you need **four times** the data. To cut it to a tenth you need a hundred times the data. Collecting more data always helps, but each extra observation helps less than the last.

Likewise, **more variability makes the SE larger**: doubling σ doubles the SE for the same n. Better measurement, or a more homogeneous population, reduces σ and is another route to a smaller SE.

5. SE for other statistics

Statistic	Standard error	Notes
Mean $\bar{x}$	$\sigma/\sqrt{n}$ (estimate: $s/\sqrt{n}$)	Independent observations
Proportion $\hat{p}$	$\sqrt{(p(1-p)/n)}$ (estimate: $\sqrt{(\hat{p}(1-\hat{p})/n)}$)	np and $n(1-p)$ both ≥ 10
Difference of two means $\bar{x}_1 - \bar{x}_2$	$\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$ (estimate: $\sqrt{(s_1^2/n_1 + s_2^2/n_2)}$)	Independent samples; variances add, never subtract
Difference of two proportions	$\sqrt{(p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2)}$	Independent samples
Mean of paired differences	$s_d/\sqrt{n}$	Treat the n differences as one sample

WORKED EXAMPLE **A poll**

A poll of n = 1,000 finds 52% support. $SE(\hat{p}) = \sqrt{(0.52 \times 0.48/1000)} = \mathbf{0.0158}$, about 1.6 percentage points. A result of 52% from this poll is well within a couple of SEs of 50%, so it does not rule out a tie.

WORKED EXAMPLE **Difference of two means**

Group A: n = 60, s = 1.8. Group B: n = 50, s = 2.4. SE of the difference $= \sqrt{(1.8^2/60 + 2.4^2/50)} = \sqrt{(0.0540 + 0.1152)} = \mathbf{0.4113}$.

COMMON MISTAKE **Variances add**

The variance of a **difference** of independent estimates is the **sum** of their variances, because uncertainty from either sample adds. Never subtract the SEs.

6. The bootstrap: estimating SE without a formula

For statistics with no simple formula (a median, a ratio, a correlation) we can **resample from our own sample**. The **bootstrap** treats the sample as a stand-in for the population: draw n values *with replacement* from the sample, compute the statistic, and repeat thousands of times. The standard deviation of those results is the **bootstrap SE**.

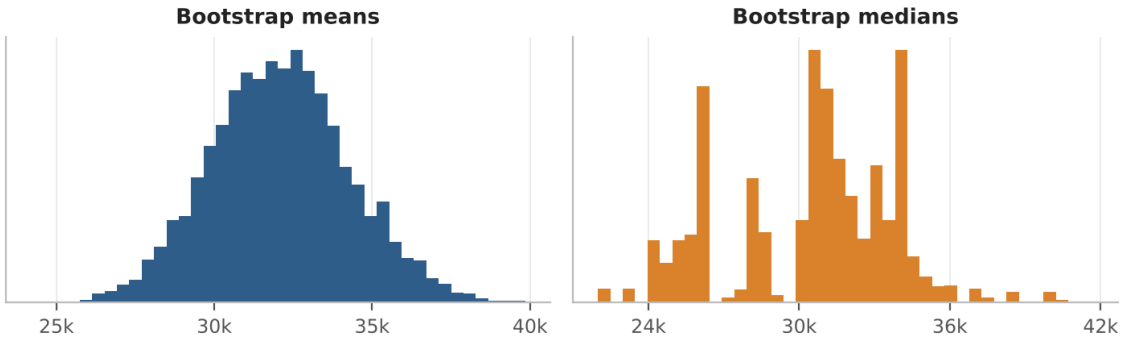


Figure 2. Bootstrap distributions of the mean and the median of one sample of 40 incomes.

Statistic	Formula SE	Bootstrap SE
Mean	s/√n = 2,218	2,216
Median	(no simple formula)	3,384

For the mean, the bootstrap SE agrees with s/√n, which builds confidence in the method. For the median, it gives an answer where the textbook has none.

7. SE, margin of error and intervals

Multiplying the SE by a critical value turns it into a **margin of error**: margin of error = reliability factor × SE (for 95% and σ known, 1.96 × SE). That margin, added to and subtracted from the estimate, is the **confidence interval** of Topic 09.

$$\text{estimate} \pm (\text{reliability factor}) \times \text{SE}$$

8. Reporting

- Say which one you are reporting: 'mean ± SD' describes the data; 'mean ± SE' describes the precision of the mean.
- On a chart, error bars of ±1 SE give only about 68% coverage; **95% confidence intervals** are usually more informative.
- Never report an SE without n.
- SE measures **random** error only. It cannot detect bias from a bad sampling method (Topic 01).

9. In Python

```
import numpy as np
from scipy import stats

x = np.array([12.1, 14.3, 11.8, 15.2, 13.7, 12.9, 14.8, 13.1])
se = x.std(ddof=1) / np.sqrt(len(x))        # manual
print(se, stats.sem(x))                     # scipy's built-in agrees

# bootstrap SE of the median
rng = np.random.default_rng(0)
meds = [np.median(rng.choice(x, len(x))) for _ in range(5000)]
print(np.std(meds))
```

SUMMARY

- SE = the standard deviation of a statistic's sampling distribution; for the mean $\sigma/\sqrt{n}$, estimated by $s/\sqrt{n}$.
- SD describes the data and stays put as n grows; SE describes the estimate and shrinks like $1/\sqrt{n}$.
- Quadruple the sample to halve the SE.
- SE for $\hat{p}$ is $\sqrt{p(1-p)/n}$; for differences, variances add.
- The bootstrap approximates SE by resampling when no formula exists.

Practice questions

Try each question before reading the answer underneath it.

Q1. $\sigma = 20$, $n = 100$. Find SE of the mean.

Answer. $20/\sqrt{100} = 2$.

Q2. To reduce that SE from 2 to 1, what sample size is needed?

Answer. The SE halves, so n must quadruple to 400.

Q3. A sample of 36 has $s = 9$. Estimate the SE of $\bar{x}$.

Answer. $9/\sqrt{36} = 1.5$.

Q4. Poll: 640 of 1,000 favour a proposal. SE of $\hat{p}$?

Answer. $\hat{p} = 0.64$; $\sqrt{(0.64 \times 0.36/1000)} = 0.0152$.

Q5. Two independent groups have $SE_1 = 3$ and $SE_2 = 4$. What is the SE of the difference of their means?

Answer. $\sqrt{(3^2 + 4^2)} = 5$ (not 7, and not 1).

Q6. Does a larger sample reduce the standard deviation of the data?

Answer. No. s settles near σ. The larger sample reduces the standard error of the mean.